

KARTHIK MANDA

Singapore PR | +65 8779 4985 | mkarthik1725215@gmail.com | [GitHub](#) | [Portfolio](#) | [Linkedin](#)

EDUCATION

Nanyang Technological University, Singapore

08/2024 – Present

COLLEGE OF COMPUTING AND DATA SCIENCE

Bachelor of Engineering (Hons), Computer Engineering

- **Relevant Coursework:** Data Structures and Algorithms, C and C++ Programming, Microprocessor System Design and Development, Object Oriented Programming, Software Engineering, Signals Systems and Transforms, Operating systems,

PROFESSIONAL EXPERIENCE

Wee Kim Wee School Of Communication and Information

02/2026 – 03/2026

Web Development Intern (Remote)

- Architected and deployed a **Conspiracy Theory Datasite** under the University Work Study Programme, managing the entire **Software Development Life Cycle (SDLC)**.
- **Collaborated with faculty and stakeholders** to translate communication requirements to high fidelity and responsive UI/UX adhering to institutional branding.
- Engineered a **modern and futuristic front-end design**, streamlined data visualization and content hierarchy enabling user friendly workflow to perform independent content updates.

PROJECTS

CodeRecon (Python, LLM, AST)

01/2026 – Present

- Engineered a **Modular Static Analysis (MSA) Engine** utilizing python's ast module to perform high density repository/codebase reconnaissance and structural mapping.
- **Developed a local first inference pipeline** using Ollama and Llama3 to generate architectural edits and refactor suggestions while maintaining zero-cloud data privacy
- **Implemented an automated CI/CD pipeline** via GitHub Actions to manage the full software distribution lifecycle, including building and **publishing releases to PyPI**.

HelioDesk (React, JavaScript, PostgreSQL)

12/2025 – 01/2026

- Developed a **personal productivity dashboard application** to manage multitasking and shortcuts access.
- Implemented cross-platform storage access with the help of **Supabase** which reduces shortcut and document access time by 25%.
- Applied **ESLint** and various other production build workflows and documented the system architecture process.

HACKATHONS & ACHIEVEMENTS

Apple Swift Student Challenge 2026

02/2026 – 03/2026

Winner, Global (Top 300)

- Selected by **Apple** from thousands of global applicants for **ArchLab**, a high performance system design simulator.
- Recognized for technical excellence in **Swift 6** concurrency, a real-time simulation logic, and UI/UX innovation.
- Awarded from bridging the gap between academic theory and production grade software architecture.

HTX HackX 2025

09/2025 – 11/2025

Finalist, (Top 15)

- Developed a **forecasting tool** to identify high density “blind spots” in Singapore’s nature trails and parks, filling a critical data gap in national hazardous chemical plume response planning.
- Led the design and development of the final incident command dashboard, translating complex predictive crowd datasets into **actionable UI visualizations for Singapore Civil Defence Force (SCDF)**.
- Selected as Top 15 Finalists to pitch our solution on the pitching day out of 60 shortlisted teams

LEADERSHIP & INVOLVEMENT

Students of Computing and Data Science Club at NTU

09/2024 – 09/2025

Program Planner - Singapore, SG

- Organised and contributed to **TechFest 2025**, a hackathon along with a career fair involving 300+ attendees, managed 6+ events over the period of 1 year with helping the club boost its participants by 20% and also supervised a team of 6 people involving event planning, ideations and onboarding companies.

INTERESTS & SKILLS

Technical: Python, Java, C, C++, Verilog, JavaScript (React, Vanilla.js), Git, HTML/CSS, Tailwind, Swift.

Tools: VS Code, GitHub, Netlify, Railway, Supabase, Figma, Adobe Photoshop, Canva.

Skills: Software development, Local LLM Orchestration, Context Engineering, Web Design, Software Project Management, Event Planning, Product Development, Model Context Protocol (MCP).

Languages: Proficient in English, Hindi and Telugu.